



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO RRD ENERGY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

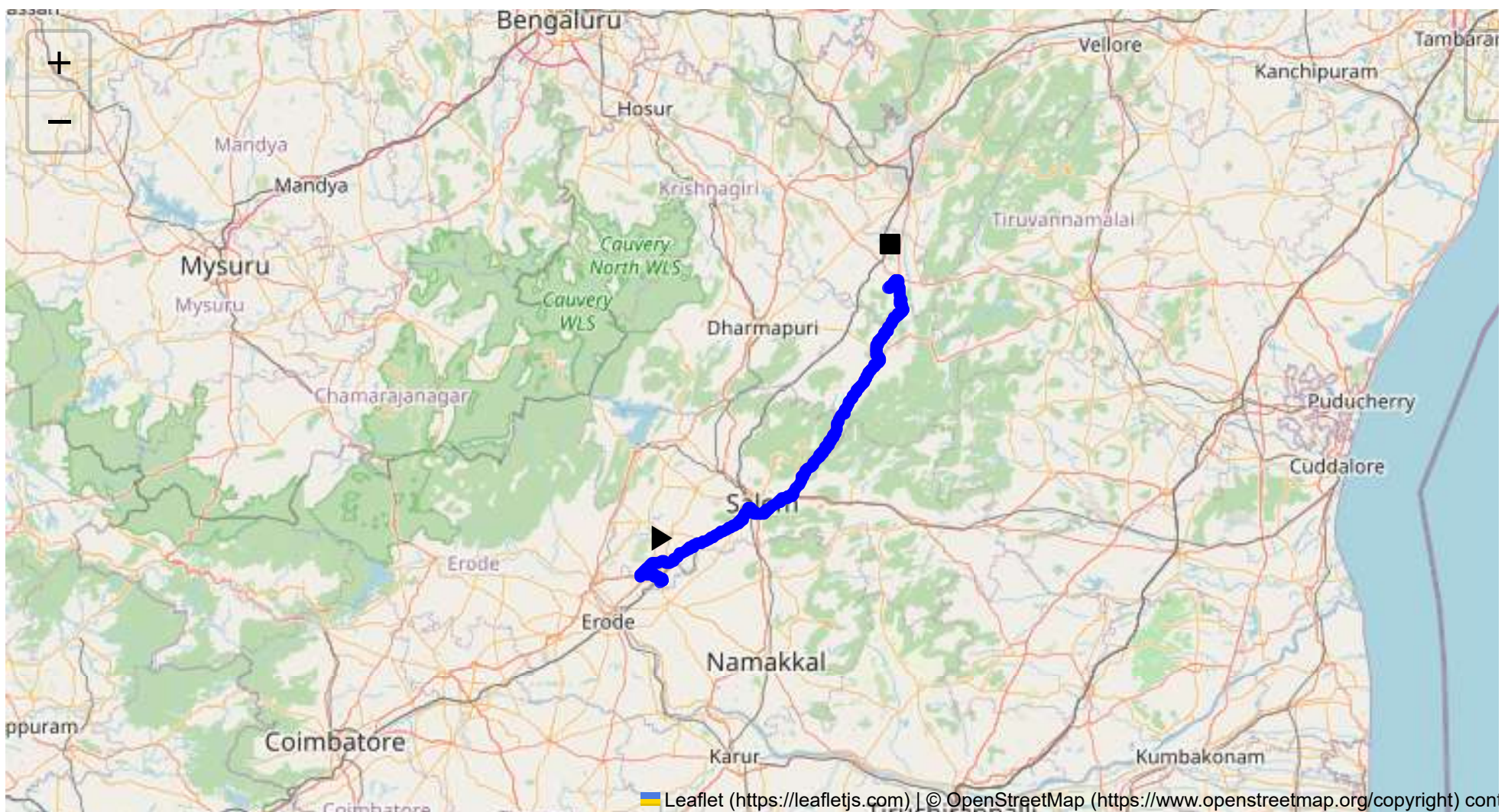
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 147.60 km
Estimated Duration: 3.4 hours
Adjusted Duration (Heavy Vehicle): 4.2 hours
Start: (11.4381, 77.8734)
End: (12.245803, 78.511026)



Welcome to the Journey Risk Management Study

Route Overview

The route spans approximately 147.60 kilometers and takes roughly 3.36 hours for heavy vehicles carrying hazardous materials. It traverses several regions in Tamil Nadu, India, connecting Sangagiri to Udam Pallam via major waypoints and highways including Bhavani Main Road and Salem – Kochi Highway.

Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu has a tropical climate. The region receives most of its rainfall from the Northeast monsoon between October and December, which could lead to slippery roads and reduced visibility. The summer months from March to June can be exceptionally hot, posing risks of tire blowouts. During the monsoon, flash floods can occur, so precautions should be taken.

Traffic Patterns and Congestion

- Peak Hours:** Typically, peak hours are from 8 AM to 10 AM and from 5 PM to 8 PM.
- Congestion-Prone Areas:**
 - Kondalampatti Bye Pass:** This area can have heavy traffic due to its proximity to Salem city.
 - Bhavani Main Road:** Known for congestion, especially during peak hours.
 - Salem City Exits:** Expect heavy traffic, which could delay transport.

Assessment of Road Quality and Infrastructure

- Highways:** Generally, highways like Salem–Kochi have good conditions. However, potholes can occur due to heavy rain.
- Rural Roads:** Some segments might have uneven surfaces and narrower roads.
- Bridges and Flyovers:** Check the latest local advisories as these can be prone to closures or damage during heavy rains.

Alternative Route Suggestions

- Via NH544 to NH44:** In case of major blockages, an alternative would be to use the NH544, connecting to NH44, though this may increase travel time.
- Local Bypasses:** Use Kondalampatti and other local bypasses if urban congestion is significant.

Local Regulations on Hazardous Material Transport

- Tamil Nadu strictly regulates the transport of hazardous materials, requiring permits and specific routes.
- Follow the advisories issued by the Tamil Nadu Pollution Control Board.

Historical Incidents Involving Heavy Vehicles

- Frequent Incidents:** Notably near Salem due to urban congestion and dense traffic.
- Heavy Rains:** Previous incidents have occurred during monsoon seasons resulting in road closures.

Environmental Considerations and Sensitive Areas

- Protected Areas:** Watch for wildlife crossings near forested regions.

- **Pollution Controls:** Ensure materials are secured to prevent spillage near water bodies, which are common along the route.

Communication Coverage

- Generally good in urban and suburban areas.
- **Dead Zones:** May exist in rural stretches; drivers should ensure GPS/maps are available offline.

Estimated Emergency Response Times

- **Sangagiri and Salem:** Response time can range from 15-30 minutes.
- **Remote Areas (e.g., Uthangarai):** Could be over 45 minutes depending on time of day and weather conditions.

Overall Summary of Risk Assessment

Transport along this route requires careful planning due to congestion in Salem and risk of weather-related hazards. Adequate preparation for communication needs and timely adherence to regulations and advisories will mitigate most risks. Transport firms should prepare emergency plans for key segments prone to delays or with lower accessibility to emergency services.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45764, 77.85219	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45545, 77.81812	30 KM/Hr
8	Turn	High	11.45584, 77.81695	15 KM/Hr
9	Turn	Medium	11.45656, 77.81709	30 KM/Hr
10	Turn	Medium	11.53569, 77.95787	30 KM/Hr
11	Turn	Medium	11.64120, 78.11797	30 KM/Hr
12	Turn	High	11.64216, 78.11875	15 KM/Hr
13	Turn	High	11.64162, 78.12160	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
14	Turn	Medium	11.64087, 78.12157	30 KM/Hr
15	Turn	High	11.63948, 78.12081	15 KM/Hr
16	Turn	Medium	11.62821, 78.13348	30 KM/Hr
17	Turn	Medium	11.62836, 78.16483	30 KM/Hr
18	Turn	High	11.66380, 78.20648	15 KM/Hr
19	Turn	High	11.67120, 78.23136	15 KM/Hr
20	Turn	High	11.67190, 78.23144	15 KM/Hr
21	Turn	Medium	11.67429, 78.23896	30 KM/Hr
22	Turn	Medium	11.80401, 78.33316	30 KM/Hr
23	Turn	Medium	11.81208, 78.34071	30 KM/Hr
24	Turn	Medium	11.83518, 78.35305	30 KM/Hr
25	Turn	Medium	11.84832, 78.36115	30 KM/Hr
26	Turn	Medium	12.01522, 78.45634	30 KM/Hr
27	Turn	Medium	12.04298, 78.48358	30 KM/Hr
28	Turn	Medium	12.05452, 78.48289	30 KM/Hr
29	Turn	Medium	12.06085, 78.47749	30 KM/Hr
30	Turn	Medium	12.06944, 78.47653	30 KM/Hr
31	Turn	Medium	12.07115, 78.47725	30 KM/Hr
32	Turn	Medium	12.07456, 78.47620	30 KM/Hr
33	Turn	Medium	12.11675, 78.49771	30 KM/Hr
34	Turn	Medium	12.17723, 78.54721	30 KM/Hr
35	Turn	High	12.17864, 78.54861	15 KM/Hr
36	Turn	High	12.18629, 78.54957	15 KM/Hr
37	Turn	Medium	12.20125, 78.54379	30 KM/Hr
38	Turn	Medium	12.21097, 78.54403	30 KM/Hr
39	Turn	Medium	12.25808, 78.53388	30 KM/Hr
40	Turn	Medium	12.25907, 78.53307	30 KM/Hr
41	Turn	High	12.26264, 78.53352	15 KM/Hr
42	Turn	Medium	12.26345, 78.53306	30 KM/Hr
43	Turn	Medium	12.26447, 78.53116	30 KM/Hr
44	Turn	Medium	12.24627, 78.51770	30 KM/Hr
45	Turn	High	12.24515, 78.51609	15 KM/Hr
46	Turn	Medium	12.24611, 78.51161	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
4	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
7	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
11	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
13	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
14	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
15	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
16	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
17	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
18	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
19	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
22	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
23	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
24	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium
31	hospital	Nanjaya Hospital	12.0550888, 78.4821929	30 km/h	Medium
32	hospital	Government Hospital, Harur	12.061178, 78.4756913	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
33	hospital	Subam Hospital	12.0595128, 78.4766713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
5	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
6	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
9	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
10	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
12	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
20	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
21	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
25	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
26	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
27	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
28	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
29	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
30	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63303, 78.12713



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45545, 77.81812



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45584, 77.81695



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45656, 77.81709



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53569, 77.95787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64120, 78.11797



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64216, 78.11875



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64162, 78.12160



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64087, 78.12157



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63948, 78.12081



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62821, 78.13348



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62836, 78.16483



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66380, 78.20648



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67120, 78.23136



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67190, 78.23144



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67429, 78.23896



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.80401, 78.33316



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.81208, 78.34071



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83518, 78.35305



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84832, 78.36115



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.01522, 78.45634



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Speed Limit: 30 KM/Hr
Coordinates: 12.04298, 78.48358



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Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.05452, 78.48289



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.06085, 78.47749



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.06944, 78.47653



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.07115, 78.47725



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.07456, 78.47620



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.11675, 78.49771



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.17723, 78.54721



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.17864, 78.54861



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.18629, 78.54957



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.20125, 78.54379



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.21097, 78.54403



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25808, 78.53388



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25907, 78.53307
